535514 Reinforcement Learning

Pre-Lecture Assignment: Lecture 3

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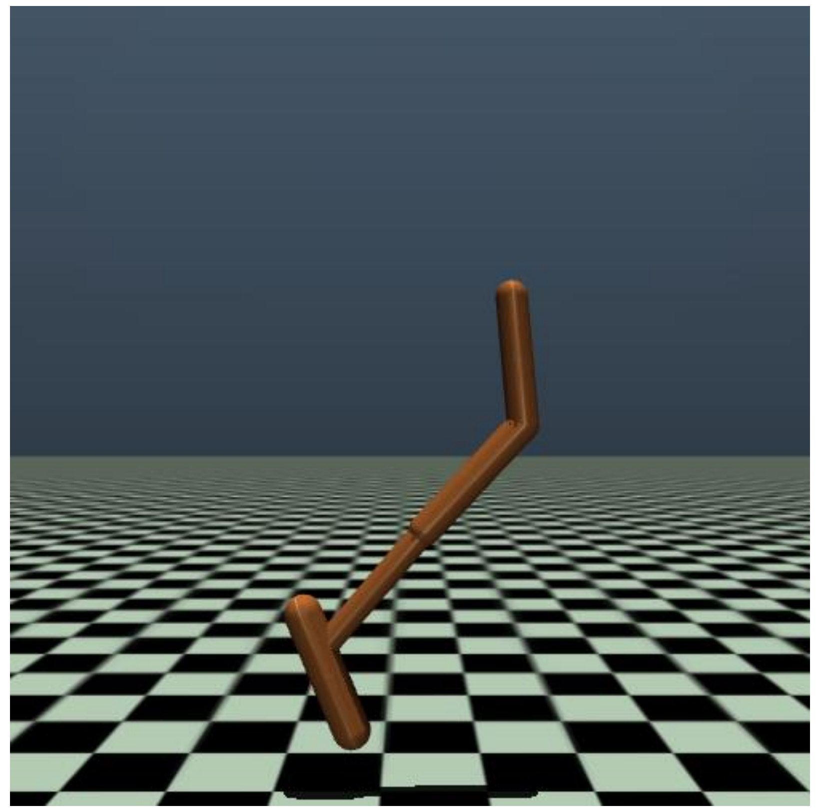
Problem: Parameterizing Policies

In Lecture 2, we learned that policy $\pi(a|s)$ is a conditional action distribution. To facilitate the learning, starting in Lecture 3, we consider parametric policies $\pi_\theta(a|s)$. You may refer to the slides **pp. 34-36** of Lecture 3 as a starting point when answering the questions below.

1) Discrete case: Suppose the state space $\mathcal{S}$ and action space $\mathcal{A}$ are both finite. How would you design the π_θ so that you can express all kinds of stationary policies?

2) Continuous case: Consider the classic RL task **Hopper**, which is a 3-DoF robot (https://gymnasium.farama.org/environments/mujoco/reacher/) with a continuous state space $\mathcal{S} = \mathbb{R}^{11}$ and an action space $\mathcal{A} = [-1,1]^3$. How would you design the π_θ so that you can express all kinds of stationary policies?



3) Constrained case: Based on 2), suppose due to some physical limitations, we need to enforce an additional action constraint as $||a||_2 \leq 1$. What changes would you make to ensure that the output actions are ***feasible*** (i.e., satisfies the constraint)?

Submission Guidelines and Remarks

- Your deliverable shall be a PDF file. Please put all your write-ups into one PDF file (photos/scanned copies are acceptable) and submit your deliverable via E3.

- If you have any questions, feel free to post them on Slack.